MATH 101 Unit 3: Polynomial, Rational, and Radical Functions - Practice 3

Version 202610122020.42 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain.

$$f(x) = 3\sqrt{x+9} - 12$$

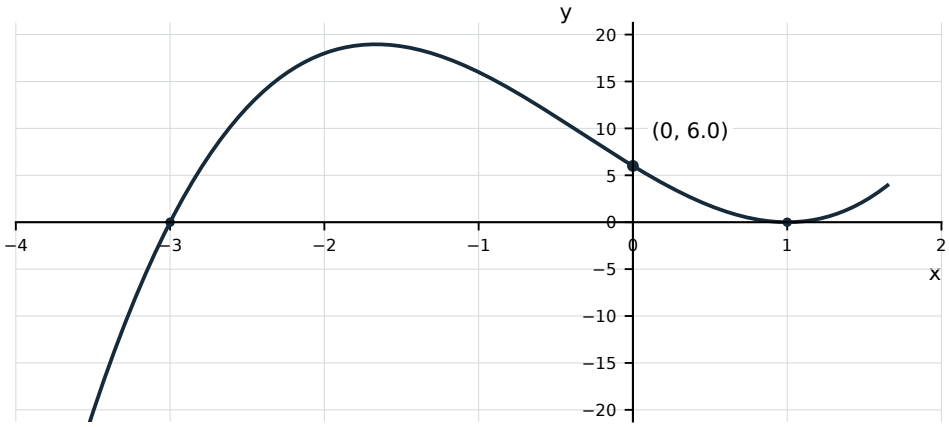
2. Find (a) the vertex and the minimum or maximum value (state which); (b) all zeros.

$$f(x) = 3(x-1)^2 - 48$$

3. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts.

$$f(x) = \frac{(x-3)(x+5)}{(x-7)(x-3)}$$

4. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions.



5. Find all zeros: (a) list Rational Root Theorem candidates, then test only 1 and -1; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros.

$$p(x) = x^3 - 4x + 3$$

Student answer sheet

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Question 1

(a) Domain

4 pts

(a) Range

4 pts

(b) x-intercept

3 pts

(b) y-intercept

3 pts

(c) Infinite end

4 pts

(c) Finite end

2 pts

Question 2

(a) Vertex

6 pts

(a) Minimum

4 pts

(b) Zeros

10 pts

Student answer sheet

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Question 3

(a) Domain — 4 pts

(b) Hole — 4 pts

(b) Vertical — 3 pts

(b) Horizontal — 3 pts

(c) x-intercept — 3 pts

(c) y-intercept — 3 pts

Question 4

Polynomial — 16 pts

Left end — 2 pts

Right end — 2 pts

Question 5

(a) Candidates/root — 5 pts

(b) Quotient — 7 pts

(c) Other zeros — 8 pts

Answer key

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Question 1

(a) Domain	$[-9, \infty)$	4 pts
(a) Range	$[-12, \infty)$	4 pts
(b) x-intercept	$(7,\ 0)$	3 pts
(b) y-intercept	$(0,\ -3)$	3 pts
(c) Infinite end	$x \to \infty:\ f(x) \to \infty$	4 pts
(c) Finite end	$x \to -9^{+}:\ f(x) \to -12$	2 pts

Question 2

(a) Vertex	$(1,\ -48)$	6 pts
(a) Minimum	-48	4 pts
(b) Zeros	$x = -3,\ 5$	10 pts

Answer key

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Question 3

(a) Domain	$\mathbb{R} \setminus \{3, 7\}$	4 pts
(b) Hole	$(3,\ -2)$	4 pts
(b) Vertical	$x = 7$	3 pts
(b) Horizontal	$y = 1$	3 pts
(c) x-intercept	$(-5,\ 0)$	3 pts
(c) y-intercept	$(0,\ -\frac{5}{7})$	3 pts

Question 4

Polynomial	$2(x+3)(x-1)^2$	16 pts
Left end	$x \to -\infty:\ f(x) \to -\infty$	2 pts
Right end	$x \to \infty:\ f(x) \to \infty$	2 pts

Question 5

(a) Candidates/root	$\{-3,\ -1, 1, 3\};\ x = 1$	5 pts
(b) Quotient	$x^2 + x - 3;\ R = 0$	7 pts
(c) Other zeros	$x = -\frac{1}{2} + \frac{\sqrt{13}}{2},\ \ -\frac{\sqrt{13}}{2} - \frac{1}{2}$	8 pts